# CAMPUS SENTINEL

*Comprehensive Technical Report, Codebase Audit & Strategic AI Roadmap*

Repository: **d:\CampusSentinal** | Enterprise NVR & AI Proctoring System | October 2026

## 1. Executive Summary & System Overview

**Campus Sentinel** is an enterprise-oriented computer vision Network Video Recorder (NVR) and automated surveillance system specifically engineered for academic institutions. The system replaces conventional passive security monitors with active AI agents capable of continuous person tracking, real-time crowd occupancy counting, polygon-bounded restricted zone (ROI) intrusion detection, and automated exam proctoring with biomechanical malpractice detection.

By leveraging deep learning models (YOLOv8 & YOLOv8-Pose) synchronized with multi-process video ingest and asynchronous WebSocket telemetry, Campus Sentinel detects illicit behaviors in sub-second latency, logs evidence directly into an SQLite audit trail, and automatically packages synchronized 15-second MP4 evidence clips and high-resolution snapshots for review.

## 2. Full Codebase & File Inventory

Structured breakdown of all files, models, configuration files, and persistence storage within the repository:

| File / Component | Type / Size | Primary Functional Role |
|---|---|---|
| run.py | Python 8.0 KB | Master entry point; coordinates FastAPI, WebSocket broadcasting, MJPEG streaming, and multi-process camera workers. |
| core/inference.py | Python 16.7 KB | Core CV engine; CameraPipeline with decoupled AI inference, ByteTrack, 17-keypoint pose heuristics, and ring buffers. |
| core/config_manager.py | Python 2.3 KB | Configuration manager; reads/writes camera entries, operating modes (CLASSROOM vs EXAM), and ROI coordinates in JSON. |
| core/event_manager.py | Python 2.5 KB | Audit logging service; initializes SQLite schema, records incidents, writes JPEG snapshots, and links video evidence clips. |
| ui/templates/index.html | HTML/JS/CSS 22.6 KB | Cyber-dark glassmorphism enterprise dashboard with live multi-camera grid, scrubber, 1x/2x/4x speed controls, and incident modal. |
| storage/config.json | JSON Config 781 B | Persistent camera registry holding live feeds, webcam indices, IP RTSP URLs, modes, and polygon boundary arrays. |
| storage/db/campus.sqlite3 | SQLite DB 20.5 KB | Relational database maintaining timestamped incident records, event types, camera IDs, snapshot paths, and clip paths. |
| storage/clips/ | Directory | Repository for saved MP4 incident video recordings (both automated 15-second incident clips and manual 2-minute captures). |
| storage/screenshots/ | Directory | Repository for high-resolution JPEG snapshots captured at the exact moment an anomaly is detected. |
| storage/model school-Channel 50.MP4 | Test Video 85.2 MB | Pre-recorded CCTV classroom/exam monitoring video used for test benching and verification. |
| yolov8n.pt | Model 6.5 MB | Ultralytics YOLOv8 Nano object detection weights; utilized in CLASSROOM mode for person detection & ROI intrusion. |
| yolov8s.pt | Model 22.6 MB | Ultralytics YOLOv8 Small object detection weights; higher accuracy fallback for object detection. |
| yolov8n-pose.pt | Model 6.8 MB | Ultralytics YOLOv8 Nano Pose weights; detects 17 skeletal landmarks per individual in EXAM mode. |
| yolov8s-pose.pt | Model 23.5 MB | Ultralytics YOLOv8 Small Pose weights; enhanced keypoint localization for crowded or distant classrooms. |

| | | |
|---|---|---|
| main.py | Python 3.7 KB | Standalone OpenCV scale-testing prototype evaluating multi-student head-turn heuristics via BoT-SORT. |
| detailed skeletal.txt | Text 4.9 KB | Early prototype design notes analyzing MediaPipe Pose, IoU collision for fight detection, and wrist extension reach. |
| readme.txt | Text 175 B | Deployment prerequisite notes (pip install fastapi uvicorn websockets ultralytics). |

# 3. Core Architecture & Workflow Analysis

## 3.1 Multiprocessing & Low-Latency Streaming Pipeline

Video surveillance architectures frequently suffer from Python's Global Interpreter Lock (GIL) and frame serialization lag. Campus Sentinel solves this using a two-tier concurrency model:

- **Dedicated Worker Process:** Each camera stream is spawned as an independent multiprocessing.Process (isolated memory space, separate core execution).
- **Decoupled IO & AI Loop:** Inside each process, an Inference Thread continuously runs YOLOv8 tracking and pose analysis, while a Main Output Thread reads frames from cv2.VideoCapture at native camera frame-rate, paints bounding boxes, manages playback speed, and pushes frames to the queue.
- **Zero-Latency Frame Dropping:** The inter-process queue between the worker and FastAPI has maxsize=1. If FastAPI is slow to serve a web client, the queue automatically discards stale frames, guaranteeing zero buffer accumulation.
- **Rolling Memory Ring Buffer:** Frames are stored in a circular collections.deque(maxlen=900), retaining the preceding 30 to 60 seconds of video in memory without writing to disk continuously.

## 3.2 Surveillance & Proctoring Modes

- **CLASSROOM Mode:** Employs yolov8n.pt with ByteTrack (bytetrack.yaml). Evaluates the bottom-center coordinates (feet) of each bounding box against custom polygon vertices via cv2.pointPolygonTest. If a person enters a restricted zone, an alert is triggered immediately.
- **EXAM Mode:** Employs yolov8n-pose.pt to track 17 anatomical keypoints per person simultaneously in a single forward pass. Implements a 3-tier biomechanical heuristic: (1) Severe Profile View (looking 90° sideways to a neighbor), (2) Facing Backwards (180° turn with hidden facial features but visible shoulders), and (3) Moderate Head Turn (yaw deviation calculated via nose-to-ear ratios exceeding ±45°).
- **Temporal Persistence Buffer & Auto-Evidence Packager:** Anomalies must persist for at least 30 consecutive frames (~1.5–2.0s). Normal postures decay the suspicion counter at double speed (-2/frame). Upon confirmation, the system automatically pulls the preceding 450 frames (15s) from memory, encodes an MP4 clip, snaps a high-res image, and notifies proctors over WebSockets.

# 4. Current Architectural & Algorithmic Limitations

- **Rule-Based Geometrics vs Action Understanding:** Current pose rules detect head rotation angles, but cannot detect micro-actions like looking down at concealed cheat sheets, smartphone reflections, whispering, or passing notes under desks.
- **MJPEG Streaming Bandwidth:** Server-Sent MJPEG multipart streams consume significant network bandwidth (~1.5–3 MB/s per camera stream), which does not scale gracefully to 50+ simultaneous campus feeds.
- **Single-Node SQLite Concurrency:** SQLite is ideal for single-node deployments but lacks distributed horizontal scaling, connection pooling, and multi-tenant access control needed for campus-wide operations.
- **Absence of Cross-Camera Re-Identification (ReID):** Track IDs are local to each camera feed. When a student transitions between rooms, identity persistence is lost.
- **No Audio Intelligence:** Surveillance is currently visual-only; acoustic indicators like murmuring, paper rustling, and verbal collusion are unmonitored.

# 5. Strategic AI Roadmap: The Part to Go Next Using AI

This roadmap details high-impact AI, Computer Vision, and Generative AI pathways to evolve Campus Sentinel into a state-of-the-art autonomous campus intelligence platform.

## Phase 1: Advanced Action Recognition & Multimodal Malpractice Detection

- **Spatio-Temporal Action Recognition (ST-GCN / VideoMAE):** Integrate Graph Convolutional Networks or VideoMAE to classify sequential human motion patterns rather than relying on static frame yaw angles. Accurately identifies passing objects, desk tampering, and concealed phone usage.
- **3D Gaze Vector Tracking (L2CS-Net / ETH-XGaze):** Deploy lightweight gaze estimation models to estimate exact 3D gaze vectors from the eye region. Differentiates between looking down at the exam paper versus glancing at a neighbor's answer sheet.
- **Zero-Shot Prohibited Object Detection (YOLO-World):** Leverage open-vocabulary object detectors to identify unauthorized items (smartphones, smartwatches, cheat notes, ear-pieces) without retraining custom detectors.

## Phase 2: Intelligent Audio Surveillance & Whispering Analysis

- **Acoustic Malpractice & Whispering Detection:** Deploy lightweight audio classification models (YAMNet / Whisper / Audio Spectrogram Transformer) on edge microphone streams to detect covert whispering, paper rustling, or commotion in exam halls.
- **Audio-Visual Sensor Fusion:** Correlate audio amplitude spikes and classified speech events with visual head-turn clusters to produce dual-evidence incident confidence scores.

## Phase 3: Cross-Camera Re-Identification (ReID) & Campus Safety

- **Cross-Camera Person Re-Identification (FastReID / OSNet):** Implement deep feature embedding extractors to track persons of interest across multiple cameras without relying on facial recognition, preserving student privacy while mapping movement trajectories.
- **Biometric Identity Verification:** Provide optional opt-in facial recognition (InsightFace / ArcFace) at exam hall entry portals to verify student identities against university registration rosters and prevent exam impersonation.

## Phase 4: Scalable Edge AI & Streaming Infrastructure

- **TensorRT & ONNX Quantization Acceleration:** Convert PyTorch/YOLO pipelines to NVIDIA TensorRT and ONNX Runtime with INT8/FP16 quantization. Enables 60+ FPS processing per GPU on Jetson or RTX servers.
- **WebRTC / HLS Low-Bandwidth Streaming:** Migrate from MJPEG to WebRTC (via aiortc) or low-latency HLS, cutting video streaming bandwidth by over 80% while reducing browser CPU consumption.
- **Distributed Microservice Architecture:** Decouple inference into microservices using Redis / Celery / Kafka with PostgreSQL and MinIO S3 object storage for multi-camera campus deployments.

## Phase 5: Generative AI Incident Intelligence & Conversational NVR

- **Autonomous Disciplinary Report Writer:** Feed the 15-second incident clip and metadata to a Vision-Language Model (e.g., Qwen2-VL, Gemini Flash, or Claude 3.5 Sonnet) to generate formal, human-readable disciplinary reports with timestamped evidence breakdowns.
- **Conversational Natural Language Video Search (Video-RAG):** Enable security chiefs and proctors to query campus surveillance in natural language: *'Show me all instances where a student looked behind them for more than 5 seconds in Exam Hall 2 between 10 AM and 12 PM'*.

# 6. Recommended Immediate Action Plan

- **Step 1 (Infrastructure):** Swap MJPEG streaming in run.py with WebRTC for scalable multi-camera viewing.

- **Step 2 (Computer Vision):** Upgrade pose heuristics in core/inference.py with L2CS-Net 3D gaze estimation to reduce false positives.
- **Step 3 (Object Detection):** Incorporate YOLO-World for real-time mobile phone and unauthorized device detection in exam halls.
- **Step 4 (Generative AI):** Connect an LLM pipeline to automatically generate disciplinary incident briefs when incidents are logged.